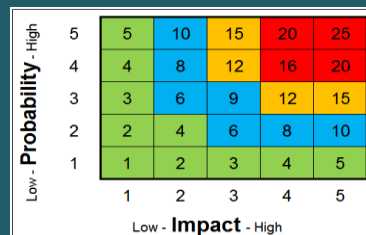


Risk Register

Project: 9802 - Asymchem - Mods to existing footprint

Last Update: 05/05/2026

Version P01



Impact score	1	No measurable impact on cost, time or scope
	3	Greater than 5% negative impact on item cost, time of delivery or scope
	5	Greater than 15% negative impact on item cost, time of delivery or scope
Probability score	1	Very low probability
	3	May occur if effective action not taking now
	5	Almost certain



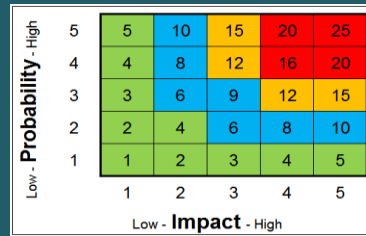
Item #	Category Cost Time Scope Quality Safety Other	Sub-category	Primary Action by	Secondary Action by	Risk Rating (Unmitigated)			Status Initiated Reviewed Actions agreed Progressing Completed	Item Description <i>There is a risk that.....</i>	Risk Rating (Mitigated)			Discussion, Mitigating Actions required/proposed	Close Out Date
					Probability (unmitigated) from 1 to 5	Impact from 1 to 5	Rating			Probability (mitigated) from 1 to 5	Impact from 1 to 5	Rating		
1	Time	Cost			3	5	15		Modifications to existing facility is not in beneficial use by Q3 2027.	2	5	10	Order long lead equipment items as early as possible Prioritise finalisation of the design of the long lead time items within FEED and place item on order	
2	Time	Cost			3	5	15		Long lead time equipment items extend the programme excessively	2	5	10	Work with supplies to understand schedule and expedite. Manage equipment deliveries to ensure that there is no programme slippage	
3	Compliance	Time			3	5	15		Amendment to current environmental permit is not granted for additional emission point on new waste tank	1	5	5	Early application and engagement with EA for buy in	
4	Quality	Technology			3	5	15		HPLC and/or Lyophilizer cannot make materials of the required quality	1	5	5	Experience inhouse from Asymchem Chinese team Similar facility is being operated in China.	
5	Cost	Technology			3	5	15		Equipment costs from China seem low and have the potential to escalate	2	5	10	Thorough discussions with suppliers to ensure equipment is specified to appropriate standards and process requirements. Equipment is CE marked so should be suitable for supply into Europe. Ensure equipment complies with UK standards and/or machine regulations Thorough review of any documentation to understand scope of supply Asymchem Chinese team have historical experience with preferred suppliers	
6	Cost	Quality of build			3	5	15		Construction of equipment supplied from China is of poor quality	1	5	5	Due diligence on manufacturers to ensure base quality of equipment FAT inspection visits to be completed Asymchem Chinese team have historical experience with preferred suppliers	
7	Cost	Scope			2	5	10		Existing utilities are not sufficient to operate the facility	1	5	5	Asymchem to have discussions with equipment suppliers to understand utilities requirements. Utilities schedule to be generated to summarise loads. Discussion in FEED phase to determine acceptable diversity	
8	Cost	Programme			2	5	10		There is no clear schedule end date	1	5	5	High level timeline provided by Asymchem. RBP to provide high level overall project programme during the feasibility study. This will be further developed during the FEED study.	
9	Cost	Operational readiness			2	4	8		Staffing for operational readiness has not been costed or allowed for	1	4	4	Training requirements to be developed during the FEED stage to enable these gaps to be addressed	
10	Cost	Technical			3	5	15		Maintenance requirements for the HPLC / Lyophilizer are not clearly defined	2	5	10	Work with supplier to understand requirements. Further discussions required during the FEED stage	
11	Cost	Strategy			3	4	12		The funding strategy is not clear for the project	3	4	12	Client has a clear funding strategy Asymchem to score	
12	Cost	Technology			3	5	15		CapEx costs are higher than expected			0	Value engineering exercise to rationalise equipment items and scope. In FEED fast track development of costs for piping and equipment installation to achieve a better accuracy. Throughout the project design lifecycle, challenge process requirements to minimise CapEx costs where possible.	
13	Time	Equipment			3	5	15		Remediation actions post FAT on equipment extending delivery dates			0	TBC	
14	Strategic	Revenue	CC		3	5	15		The cost of the project outweighs its financial benefit			0	Business case to be fleshed out throughout at each mile stone to track financial benefit.	
15	Time Cost	Technology			3	5	15		There will be scope creep in the project requirements including addition of equipment items and materials handling	1	5	5	Requirements to be frozen during early stages of FEED Continue to review the scope	
16	Cost				5	5	25		Operation of filter driers are required during demolition work and installation of Lyophilizer	4	5	20	Not able to move spray drier to alternative position as new HPLC installation in that area. Currently can schedule spray dryer operation outside of demolition work. During build phase, this risk needs to be managed as there is currently only short term visibility of operational demands. Need to understand build timelines to fully assess probability	
17	Safety				3	5	15		There is poor maintenance access around the new equipment (operability)	2	5	10	Feasibility study to include access for maintenance for the equipment Discussions with suppliers and information provided	

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18	Safety				3	5	15		There is poor access for maintenance activities on existing equipment	2	5	10	Review during feasibility study	
19	Safety				2	5	10		Egress routes and compliance to fire regulations will be compromised	1	5	5	Review layout during feasibility study and amend as necessary	
20	Quality	Operational readiness			3	5	15		The filter dryer and Lyophilizer cannot be operated at the same time as this contravenes Quality and Regulatory controls	1	5	5	No regulatory impact Initial discussions with the Quality team suggest that the existing procedural controls are adequate, and the operation of both units does not present any concerns.	
21	Cost	Operational readiness			3	5	15		The introduction of new equipment into the building cannot be achieved without modification to the existing access routes.	1	5	5	Early assessment suggests that there is a route for the new equipment to be brought into the building	
22	Cost	Strategy			3	3	9		Personal in / out Enty system - Air lock required	1	3	3	Layout can accommodate an in/out entry system	
23	Cost	Strategy			3	4	12		Fire access and egress routes	1	4	4	Survey to be completed to ensure these are compliant	
24	Compliance	Fire protection			3	4	12		Temperature within the rooms cannot be controlled adequately as there is no cooling capabilities on the current HVAC system. There is no dehumidification on the current AHU	2	4	8	Unable to confirm until heat out put from equipment is supplied from the equipment venders. Assumption that current arrangement is able to handle loads and no separate HVAC cooling package is required	
25	Cost	Technology			3	5	15		DAC arrangement does not ergonomically support normal operation and removal of resin.	2	5	10	Supplier confirms that no maintenance activities are required at the top of the column. For the feasibility study assume that the current arrangement is suitable. Further development is required at the FEED stage as there are still unknown activities that may be challenging due to heights and consideration needs to be made around manual handling of the piston end assembly and the piston.	
26	Safety	Quality of build			3	5	15		CE Certification on equipment			0	For the feasibility stage, assume no gaps	
27	Time	Equipment			3	5	15		There are no suitable tie in points for the HTF hence will have be drained down fully to enable removal of the redundant pipework.	1	5	5	Ensure cost of replacement HTF is included in the CapEx costs Impact on other equipment items using HTF and downtime need to be assessed during the FEED stage	
28	Strategic	Revenue					0		A mechanism of drying is not provided post CIP activities				Score once further information is received	
29	Strategic	Revenue					0		Purified CIP water supply does not have sufficient capacity			0	Score once further information is received	
30	Scope	Cost			3	3	9		Room balancing cannot be easily be achieved in the isolator area	1	3	3	For the feasibility study assume that isolator is running all the time	
31	Scope	Cost			3	5	15		The wall is not needed between the Lyophiliser and vacuum pumps	1	5	5	Assume that it is required for the feasibility study.	
32	Scope	Cost			3	3	9		That the vent streams cannot be routed to the existing vent header	2	3	6	Confirm composition of vent stream assess suitability of vent route	